

AIoT-Based Real-Time Solid Waste Classification Using TinyML on ESP32-S3

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Abstract—The mismanagement of municipal solid waste remains a critical environmental challenge, particularly in urban areas of developing countries where improper disposal reduces recycling efficiency and increases contamination of waste streams. Traditional computer vision solutions for waste classification rely on cloud computing and high-performance hardware, making them unsuitable for low-cost, distributed deployments.

This work presents an AIoT-based system for real-time waste classification implemented on a resource-constrained microcontroller. The proposed solution uses a Saeed Studio XIAO ESP32-S3 Sense running a TinyML convolutional neural network developed with the Edge Impulse platform. A custom dataset of 1,349 images was captured directly with the embedded OV2640 camera to minimize the sensor domain gap commonly observed when deploying models trained on external datasets. The optimized model classifies waste into four categories—recyclable, non-recyclable, organic, and idle—using post-training int8 quantization and the EON compiler to fit within the memory constraints of the microcontroller. Experimental results on a held-out test set show a classification accuracy of 98.49%, an area under the ROC curve of 1.00, a weighted F1 score of 0.99, an inference latency of 292 ms, and a peak RAM usage of only 82.8 KB. Additionally, the system integrates local LED feedback and remote monitoring through the Ubidots IoT platform via MQTT. The results demonstrate that TinyML enables efficient and scalable waste classification solutions suitable for smart city environments.

Index Terms—AIoT, TinyML, Edge Impulse, Computer Vision, Waste Classification, ESP32-S3, IoT, Convolutional Neural Network.

I. INTRODUCTION

A. Context and Justification

Global urbanization has led to a rapid increase in municipal solid waste (MSW) generation. According to the World Bank report *What a Waste 2.0*, the world generates approximately 2.01 billion tonnes of MSW annually, and this figure is projected to reach 3.40 billion tonnes by 2050 [1]. In Latin America, the situation is particularly challenging: recycling rates remain below 5% in most countries, largely because waste is not properly classified at the source. When recyclable and organic materials are mixed with general waste, recovery becomes technically difficult and economically unfeasible, resulting in the majority of MSW ending up in landfills.

Recent advances in artificial intelligence have enabled automated waste classification using computer vision tech-

niques based on deep learning. Convolutional neural networks (CNNs) have demonstrated strong performance in recognizing waste categories from images [2]. However, most existing approaches rely on cloud-based inference or high-performance computing hardware such as GPUs. These solutions require significant computational resources, stable internet connectivity, and high energy consumption, which severely limits their scalability and practicality in real-world distributed deployments.

The Artificial Intelligence of Things (AIoT) paradigm proposes an alternative approach by embedding intelligence directly into edge devices. TinyML is a subfield of machine learning focused on deploying inference models on microcontrollers with extremely constrained processing power (often below 400 MHz), limited memory (typically hundreds of kilobytes of RAM), and low energy consumption (in the milliwatt range) [3]. By performing inference locally on the device rather than in the cloud, TinyML systems achieve low latency, improved privacy, and full autonomy from network connectivity.

B. Related Work

Several studies have explored automated waste classification using deep learning. Yang and Thung [4] introduced the TrashNet dataset containing 2,527 images across six waste categories and trained a support vector machine classifier, achieving 63% accuracy. Subsequent works have improved upon this baseline using deeper CNN architectures. Shi et al. [5] proposed an enhanced deep learning framework for garbage classification that achieved over 99% accuracy on benchmark datasets, leveraging architectures with millions of parameters. Similarly, Kumar et al. [6] developed an intelligent waste sorting system for urban sustainability using large-scale neural networks with high computational requirements.

However, these high-accuracy models are typically deployed on desktop computers or cloud infrastructure and are not suitable for embedded systems. Zhu et al. [7] proposed an IoT-based smart waste management system, but relied on server-side processing for the classification task. Ruparel et al. [8] developed an AI-based smart bin, though their approach still depends on external computing resources. Kunwar [9] explored waste sorting on mobile devices, representing a step

toward edge deployment but still requiring relatively powerful hardware compared to microcontrollers.

Recent TinyML research has highlighted the feasibility of deploying optimized neural networks on microcontrollers through techniques such as model quantization, pruning, and knowledge distillation [3], [10]. Lightweight architectures like MobileNet [11] have been proposed for resource-constrained environments, but their memory footprint often exceeds the capabilities of ultra-low-power microcontrollers. The MLPerf Tiny benchmark [10] has established standardized evaluation criteria for TinyML workloads, demonstrating that meaningful inference is achievable on devices with less than 256 KB of RAM.

A critical challenge that remains underexplored in the literature is the sensor domain gap: models trained on high-quality images from public datasets often fail when deployed on embedded cameras with lower resolution, different noise characteristics, and limited optics [9]. This work directly addresses this gap by constructing a dataset captured with the same camera used during inference.

C. Contributions

This work addresses the aforementioned challenges by developing a lightweight convolutional neural network optimized for deployment on an ESP32-S3 microcontroller. The main contributions are:

- Design of a complete low-cost AIoT system for real-time waste classification using the XIAO ESP32-S3 Sense platform, with a total hardware cost below 15 USD.
- Construction of a custom dataset of 1,349 images captured directly with the embedded OV2640 camera under varied lighting conditions to eliminate the sensor-domain mismatch.
- Development and optimization of a TinyML CNN architecture achieving 98.49% test accuracy with only 82.8 KB of RAM and 292 ms inference latency.
- Quantitative comparison between int8 quantized and float32 model variants demonstrating an $8\times$ reduction in inference time with no loss in accuracy.
- Integration of edge-based inference with IoT monitoring through the Ubidots platform using MQTT communication.

II. PROPOSED SOLUTION

A. System Overview

The proposed system is designed as an intelligent assistant for waste disposal at the point of generation. The embedded device is positioned above a waste container and continuously monitors the disposal area. When a waste object is placed in the field of view, the camera captures an image that is processed locally by a TinyML model running entirely on the microcontroller. The classification result triggers immediate user feedback through colored LEDs and is simultaneously transmitted to a cloud-based IoT dashboard for remote monitoring and data logging.

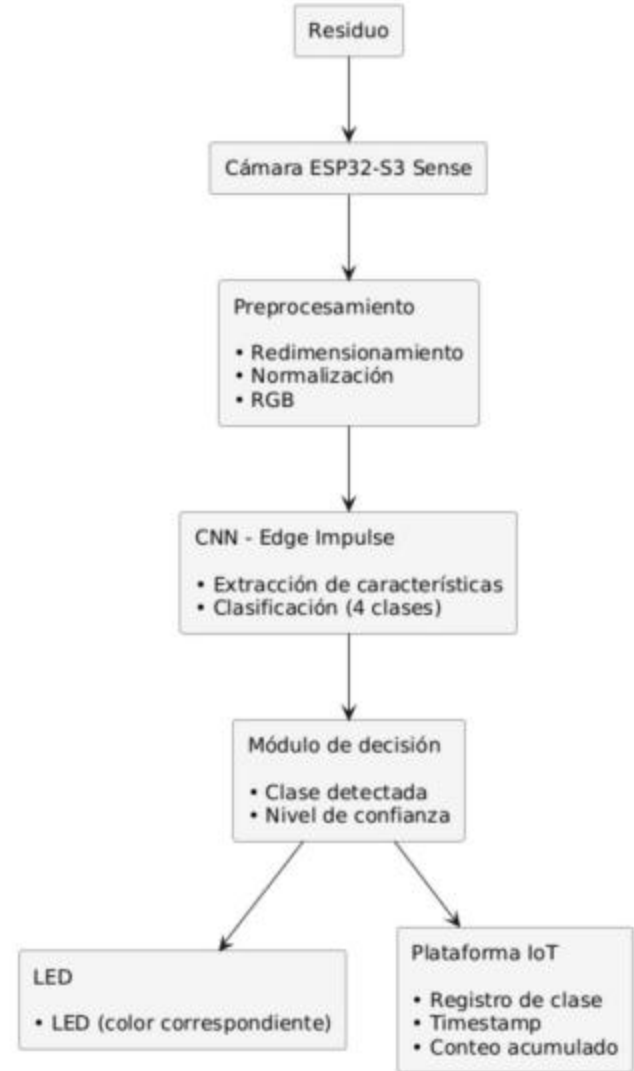


Fig. 1. System architecture showing the pipeline from image acquisition on the ESP32-S3 to local feedback and cloud monitoring through MQTT.

The end-to-end system architecture is illustrated in Fig. 1. The pipeline consists of four sequential stages: (1) image acquisition via the OV2640 camera, (2) preprocessing including resizing and pixel normalization, (3) on-device CNN inference using the quantized model, and (4) output through both local actuators and remote IoT communication.

B. Hardware Platform

The system is built around the Seeed Studio XIAO ESP32-S3 Sense, a compact development board that features the Espressif ESP32-S3 dual-core Xtensa LX7 processor clocked at 240 MHz [12]. The board provides 8 MB of PSRAM and 8 MB of flash memory, and integrates an OV2640 camera module capable of capturing images up to 1600×1200 pixels. Its compact form factor (21×17.5 mm) and low power consumption make it suitable for embedded IoT applications. The peripheral hardware includes two LEDs (green and red) for visual feedback indicating the waste category. The complete prototype assembled on a breadboard is shown in Fig. 2, where the ESP32-S3 camera module, the indicator LEDs, and a waste object under classification are visible.



Fig. 2. Prototype of the waste classification system showing the XIAO ESP32-S3 Sense with the OV2640 camera, indicator LEDs, and a non-recyclable waste item being classified.

C. Edge Impulse Workflow

The model development pipeline was implemented using the Edge Impulse platform, which provides an end-to-end workflow for building, training, and deploying TinyML models. The impulse design, illustrated in Fig. 3, defines three sequential blocks: (1) an image data block configured to accept 64×64 pixel RGB inputs (12,288 features), (2) an image processing block that normalizes pixel values, and (3) a classification learning block based on a custom CNN architecture with four output classes: recyclable, idle, non-recyclable, and organic.

D. System Modules

The system is composed of four functional modules:

- **Vision Module:** The OV2640 camera captures RGB images that are resized from VGA resolution to 64×64 pixels for inference.
- **Inference Engine:** A quantized CNN processes the image on-device and outputs classification probabilities for each waste category. The class with the highest confidence score above a configurable threshold determines the prediction.
- **Interaction Module:** Colored LEDs provide immediate visual feedback to the user: green for recyclable, red for non-recyclable, and alternating patterns for organic and idle states.
- **Communication Module:** Wi-Fi connectivity enables the ESP32-S3 to publish classification results, confidence scores, and cumulative waste counts to the Ubidots IoT platform via MQTT [13].

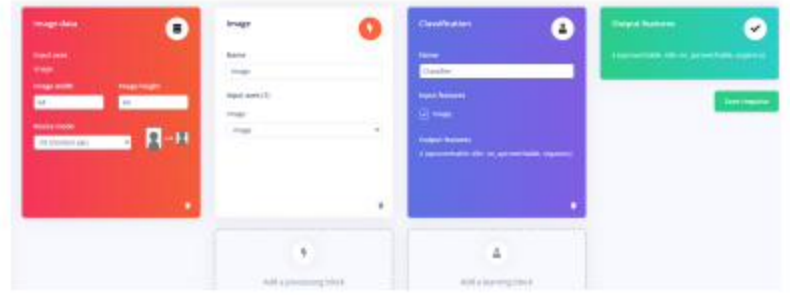


Fig. 3. Edge Impulse impulse design showing the data acquisition, image processing, and classification blocks configured for 64×64 RGB input with four output classes.

III. IMPLEMENTATION DETAILS

A. Dataset Construction

A fundamental challenge in deploying TinyML vision models is the sensor domain gap: models trained on high-resolution images from public datasets such as TrashNet [4] often exhibit significant accuracy degradation when deployed on embedded cameras with different optical characteristics, lower resolution, and distinct noise patterns. To mitigate this problem, a custom dataset was captured directly using the OV2640 camera integrated into the XIAO ESP32-S3 Sense board, ensuring that training and inference images share the same sensor characteristics.

A total of 1,349 images were collected and labeled into four classes, as detailed in Table I. The *recyclable* class includes plastic bottles, cardboard, paper, and aluminum cans. The *non-recyclable* class contains food wrappers, styrofoam, and contaminated materials. The *organic* class includes fruit peels, food scraps, and biodegradable items. The *idle* class captures the empty disposal area with no object present, which is essential for the system to distinguish between active classification events and standby states.

TABLE I
DATASET DISTRIBUTION ACROSS WASTE CATEGORIES

Class	Images	Proportion
Recyclable	354	26.24%
Non-recyclable	368	27.27%
Organic	348	25.79%
Idle	279	20.68%
Total	1,349	100%

Images were captured under three distinct lighting conditions—natural daylight, artificial LED illumination, and low-light scenarios—to improve model robustness against environmental variability. Each image was captured at VGA resolution (640×480) and stored with its corresponding class label in the Edge Impulse data acquisition module. The dataset was split using an 80/20 ratio for training and validation, with a separate test set used for final evaluation.

B. Preprocessing and Feature Extraction

The image preprocessing pipeline operates in two stages. First, captured images are resized from VGA resolution to

64×64 pixels using the “fit shortest axis” resize mode, which preserves the aspect ratio and crops the excess region. Higher resolutions such as 96×96 and 128×128 were experimentally evaluated but exceeded the available SRAM during on-device inference, causing memory allocation failures on the ESP32-S3.

Second, pixel normalization is applied to each RGB channel independently according to:

$$X_{\text{norm}} = \frac{X - \mu}{\sigma} \quad (1)$$

where X represents the raw pixel value, μ is the channel mean, and σ is the channel standard deviation computed over the training set. This normalization reduces sensitivity to illumination changes and improves convergence during training. The complete image processing block executes in 7 ms with a peak RAM usage of 4 KB on the target hardware, representing a negligible overhead compared to the classification stage.

C. CNN Architecture

The classification model was implemented as a lightweight convolutional neural network specifically designed for embedded inference within the memory constraints of the ESP32-S3. The architecture was intentionally kept shallow to minimize the number of trainable parameters while maintaining sufficient representational capacity for four-class waste discrimination.

The network architecture is detailed in Table II. The input layer receives a flattened 64×64×3 RGB tensor (12,288 features). Two convolutional blocks follow, each consisting of a 2D convolution with a 3×3 kernel and a 2×2 max-pooling layer. The first block uses 16 filters and the second uses 32 filters, progressively increasing feature abstraction while reducing spatial dimensions. After flattening, a fully connected layer with 64 neurons provides the classification embedding. A dropout layer with a rate of 0.25 is applied before the output layer to regularize the model and prevent overfitting. The final dense layer uses a softmax activation function to produce probability estimates for the four output classes.

TABLE II
CNN ARCHITECTURE SUMMARY

Layer	Output Shape	Parameters
Input	64×64×3	–
Conv2D (16 filters, 3×3)	62×62×16	448
MaxPooling2D (2×2)	31×31×16	0
Conv2D (32 filters, 3×3)	29×29×32	4,640
MaxPooling2D (2×2)	14×14×32	0
Flatten	6,272	0
Dense (64, ReLU)	64	401,472
Dropout (0.25)	64	0
Dense (4, Softmax)	4	260
Total		406,820

The model was trained using the Adam optimizer with a learning rate of 0.0005 over 30 training cycles (epochs) on a CPU-based training environment within Edge Impulse.

The total number of trainable parameters is 406,820, with the majority (98.7%) concentrated in the fully connected layer due to the large flattened feature vector of 6,272 elements.

D. Model Optimization and Deployment

To enable deployment within the strict memory constraints of the microcontroller, post-training quantization was applied to convert all 32-bit floating-point weights and activations into 8-bit integers (int8). This technique reduces the model size by approximately 4× while introducing minimal accuracy degradation.

Additionally, the Edge Impulse EON compiler was employed to further optimize memory allocation and execution speed. The EON compiler analyzes the computational graph and generates optimized C++ code tailored to the target hardware, achieving 17% less RAM usage and 23% less ROM usage compared to the standard TensorFlow Lite Micro runtime, with no loss in classification accuracy.

The final optimized model was exported as an Arduino library and integrated into the firmware of the XIAO ESP32-S3 Sense. The firmware handles the complete inference pipeline: camera initialization, image capture, preprocessing, model inference, LED control, and MQTT data transmission to the Ubidots platform.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Training Performance

The model was trained on the 80% training split and evaluated on the 20% validation split. After 30 training epochs, the model achieved a validation accuracy of 98.2% with a cross-entropy loss of 0.11. The validation confusion matrix revealed that the most challenging class was *non-recyclable*, with 95.2% correct classification (3.2% misclassified as recyclable and 1.6% as organic). The *recyclable* and *idle* classes achieved 100% validation accuracy, while *organic* reached 98.1% (with 1.9% confused as idle). The weighted average precision, recall, and F1 score on the validation set were all 0.98, with an area under the ROC curve (AUC) of 1.00.

B. Test Set Evaluation

The final model was evaluated on a separate held-out test set to assess generalization performance. The system achieved an overall test accuracy of 98.49%. Table III summarizes the key classification metrics.

TABLE III
TEST SET CLASSIFICATION METRICS

Metric	Value
Accuracy	98.49%
Area under ROC Curve	1.00
Weighted Avg. Precision	0.99
Weighted Avg. Recall	0.99
Weighted Avg. F1 Score	0.99

The per-class performance on the test set is shown in the confusion matrix (Fig. 4). The *idle* and *organic* classes

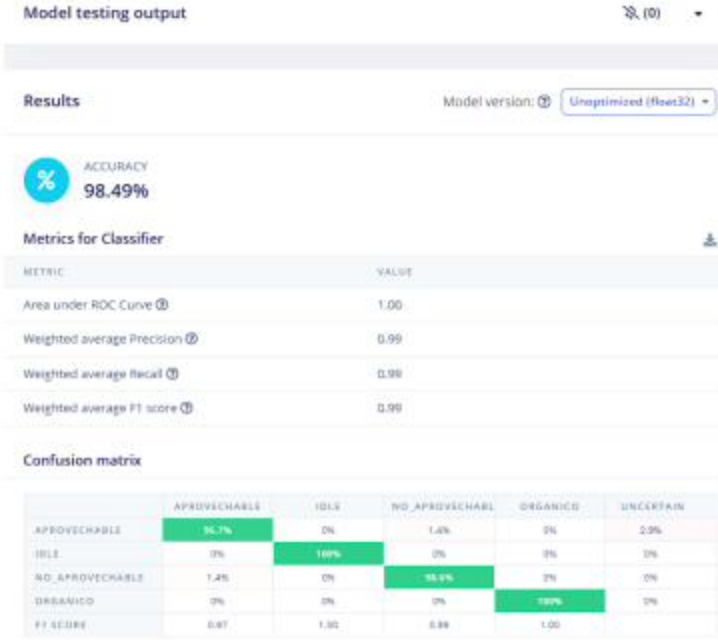


Fig. 4. Confusion matrix on the test set showing per-class classification performance across the four waste categories.

achieved perfect classification (100%). The *non-recyclable* class reached 98.6%, with only 1.4% of samples misclassified as recyclable. The *recyclable* class showed 95.7% accuracy, with 1.4% confused as non-recyclable and 2.9% classified as uncertain. Per-class F1 scores were 0.97, 1.00, 0.99, and 1.00 for recyclable, idle, non-recyclable, and organic, respectively.

The slight confusion between recyclable and non-recyclable items is expected, as certain objects such as contaminated plastic containers share visual features with both categories. Despite this, the high F1 scores across all classes confirm that the model generalizes well to unseen data captured under real-world conditions.

C. On-Device Performance and Quantization Analysis

A critical evaluation criterion for TinyML deployments is the trade-off between model accuracy and computational efficiency. Table IV presents a direct comparison between the unoptimized float32 model and the quantized int8 model deployed with the EON compiler.

TABLE IV
COMPARISON OF FLOAT32 VS. INT8 QUANTIZED MODEL

Metric	Float32	Int8	Reduction
Inference time	2,337 ms	292 ms	8.0×
Peak RAM	321.6 KB	82.8 KB	3.9×
Flash usage	169.4 KB	66.2 KB	2.6×
Accuracy	98.49%	98.49%	—

The quantized model achieves an 8 $\times$ reduction in inference latency (from 2,337 ms to 292 ms) and a 3.9 $\times$ reduction in RAM usage (from 321.6 KB to 82.8 KB), with no degradation in classification accuracy. This result demonstrates that int8 quantization is highly effective for this task and model architecture. The 292 ms total inference time (7 ms for

image preprocessing plus 285 ms for the classifier) enables responsive real-time classification suitable for interactive waste disposal scenarios.

D. IoT Monitoring

Classification results are published to the Ubidots IoT platform via MQTT over Wi-Fi. Each inference event transmits the predicted class, the confidence score, and a cumulative count per category. The IoT dashboard (Fig. 5) enables remote visualization of waste disposal patterns in real time and provides historical data that could support data-driven optimization of waste collection routes and schedules.

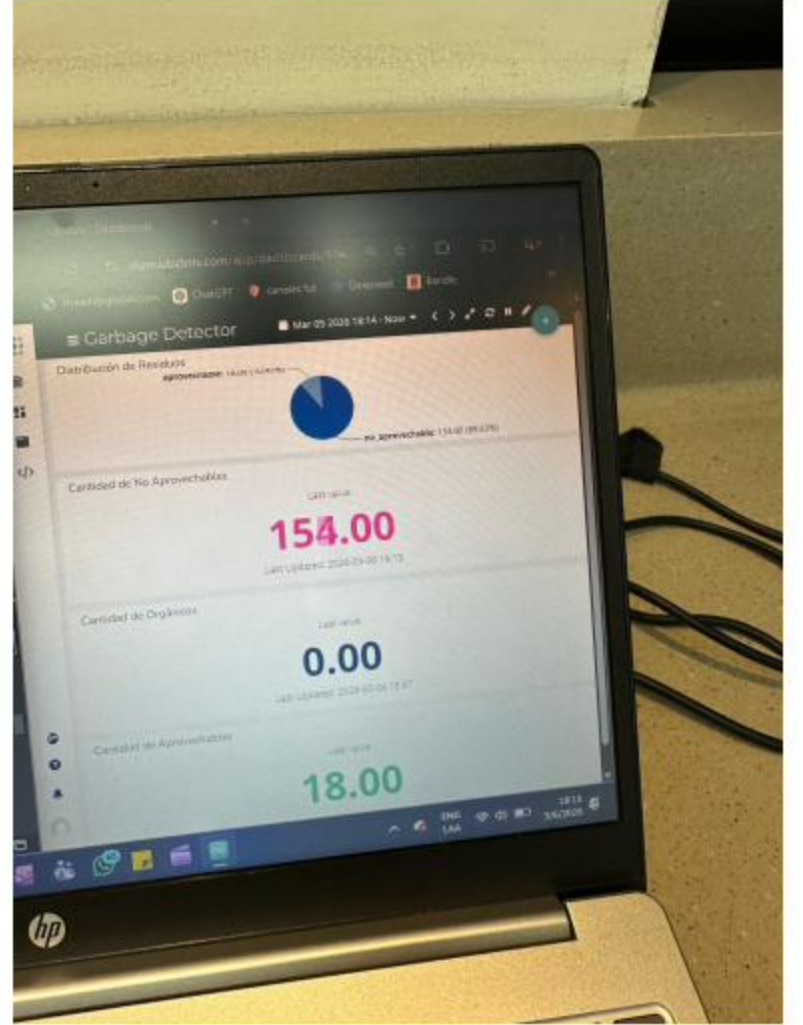


Fig. 5. Ubidots IoT dashboard displaying the waste distribution chart and cumulative classification counts per category during a live testing session.

E. Comparison with Related Work

Table V compares the proposed system with representative works in waste classification. While cloud-based approaches achieve marginally higher accuracy, they require orders of magnitude more computational resources. The proposed TinyML system achieves competitive accuracy at a fraction of the cost and energy consumption, making it suitable for scalable, distributed deployments.

TABLE V
COMPARISON WITH RELATED WORK

Work	Accuracy	Platform	Edge
Shi et al. [5]	>99%	GPU server	No
Kumar et al. [6]	>99%	GPU server	No
Kunwar [9]	92%	Mobile	Partial
This work	98.49%	ESP32-S3	Yes

F. Limitations

Although the system demonstrated high classification accuracy, several limitations should be acknowledged. First, objects must be positioned within approximately 15 cm of the camera to ensure adequate focus and image clarity, which constrains the physical design of the waste container. Second, highly reflective materials (e.g., metallic wrappers) may introduce specular glare that degrades classification performance under certain lighting conditions. Third, the system assumes that a single object is present in the field of view at a time; overlapping or multiple objects would require an object detection architecture rather than a whole-image classifier. Finally, the dataset size of 1,349 images, while sufficient for the current four-class problem, may need to be expanded for finer-grained classification tasks.

V. CONCLUSIONS AND FUTURE WORK

This work demonstrates the feasibility of implementing real-time computer vision applications on resource-constrained edge devices using TinyML techniques. A waste classification system was successfully deployed on the XIAO ESP32-S3 Sense microcontroller, achieving 98.49% accuracy on a held-out test set with a weighted F1 score of 0.99, while operating within 82.8 KB of RAM and 292 ms of inference latency.

The construction of a custom dataset captured directly with the embedded camera proved essential to bridge the sensor domain gap and achieve high classification accuracy without relying on external datasets or transfer learning from high-resolution image corpora. The application of int8 post-training quantization combined with the EON compiler enabled an 8× speedup in inference time and a 3.9× reduction in memory usage with zero accuracy loss, confirming the effectiveness of these optimization techniques for embedded deployment.

By performing all inference locally on the device, the system eliminates network dependency for the classification task, reduces latency to sub-second levels, and ensures continuous operation regardless of connectivity. The integration with the Ubidots IoT platform through MQTT provides a complementary layer for remote monitoring and data analytics without compromising the real-time performance of the edge device.

Future work will explore three main directions. First, extending the system toward lightweight object detection architectures (e.g., FOMO or YOLOv8-nano) to enable simultaneous multi-object classification. Second, investigating energy harvesting and low-power sleep modes to enable battery-powered autonomous deployments. Third, scaling the

approach across multiple distributed devices using federated learning to collaboratively improve the model without centralizing sensitive image data.

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